

Legend

Unless specified otherwise in an individual visual legend, the color coding used throughout this report follows the adjacent **PM10** and **PM2.5** concentration ranges.

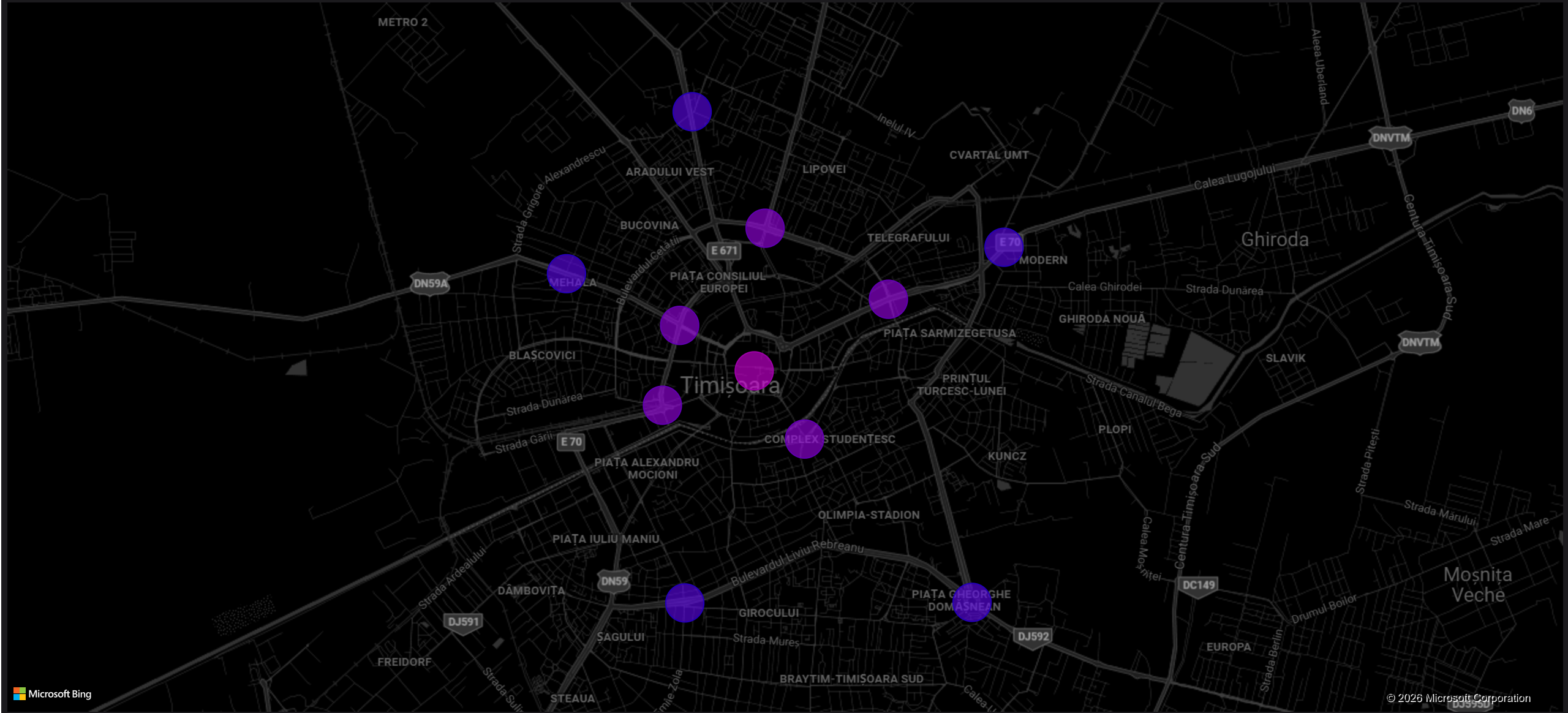
All values in this report are expressed in **micrograms per cubic meter** ($\mu\text{g}/\text{m}^3$).

These ranges are based on the **European Air Quality Index** values published by the **European Environment Agency (EEA)** and are used as a visual guide for interpreting pollution levels.

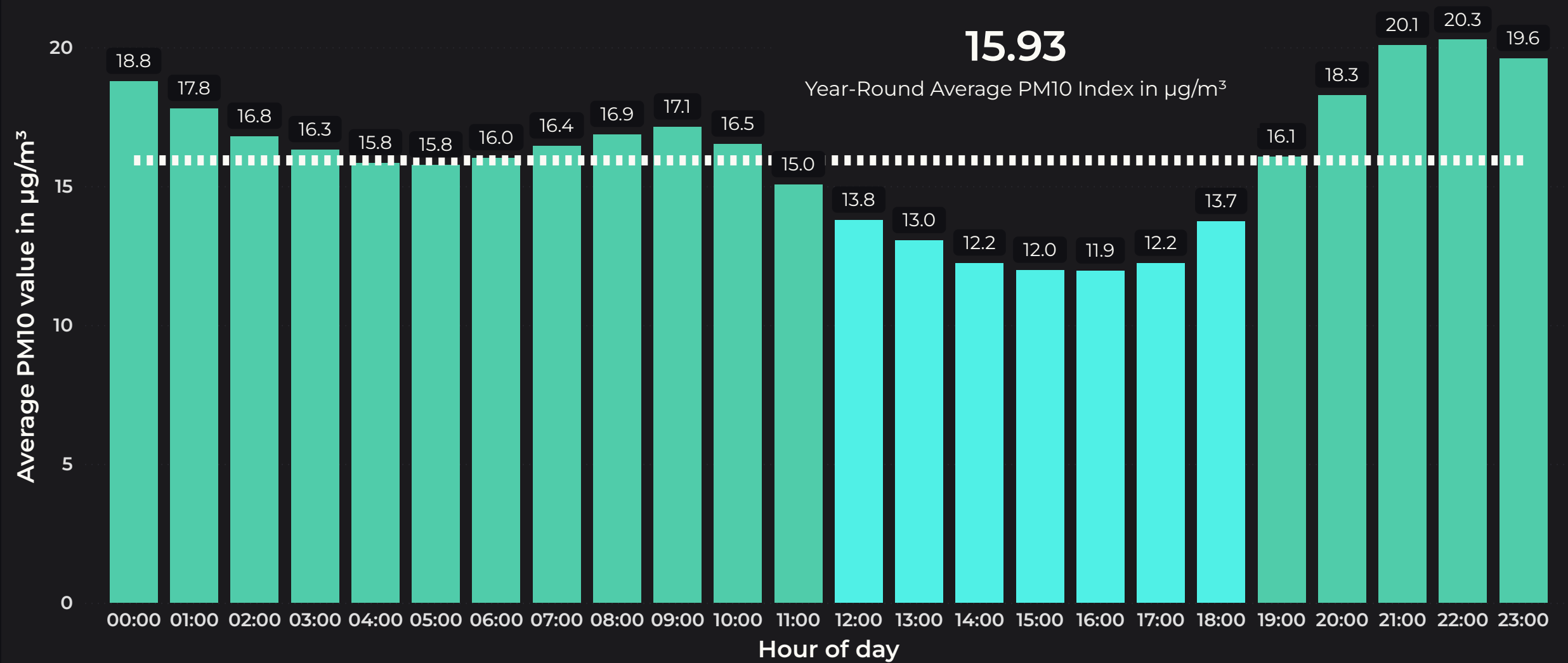
Order ▲	Level	PM10	PM2.5
1	Good	0–15	0–5
2	Fair	16–45	6–15
3	Moderate	46–120	16–50
4	Poor	121–195	51–90
5	Very poor	196–270	91–140
6	Extremely poor	271+	141+

What air quality monitoring nodes from Timisoara am I using in this report?

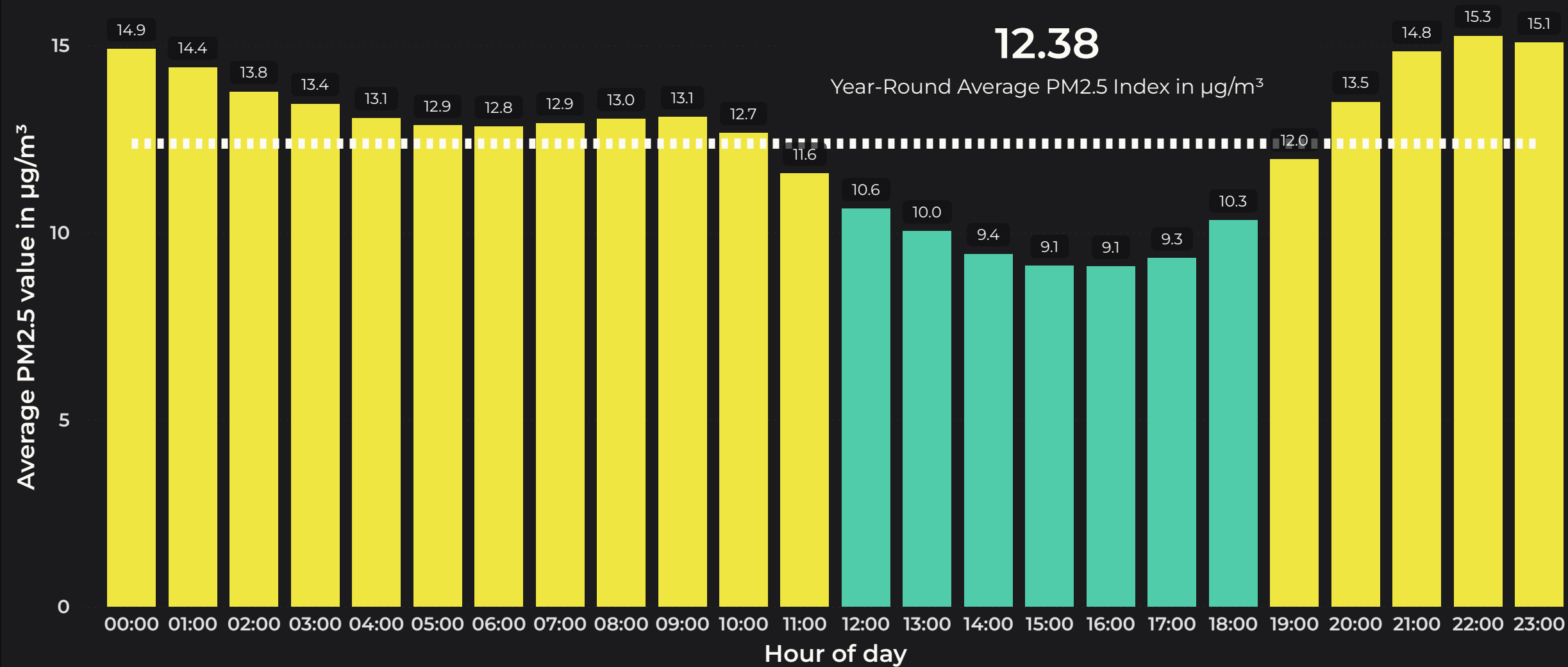
Node category based on city zone: ● Central ● Suburbs ● Ultracentral



During which hours of the day is air pollution highest throughout the city?



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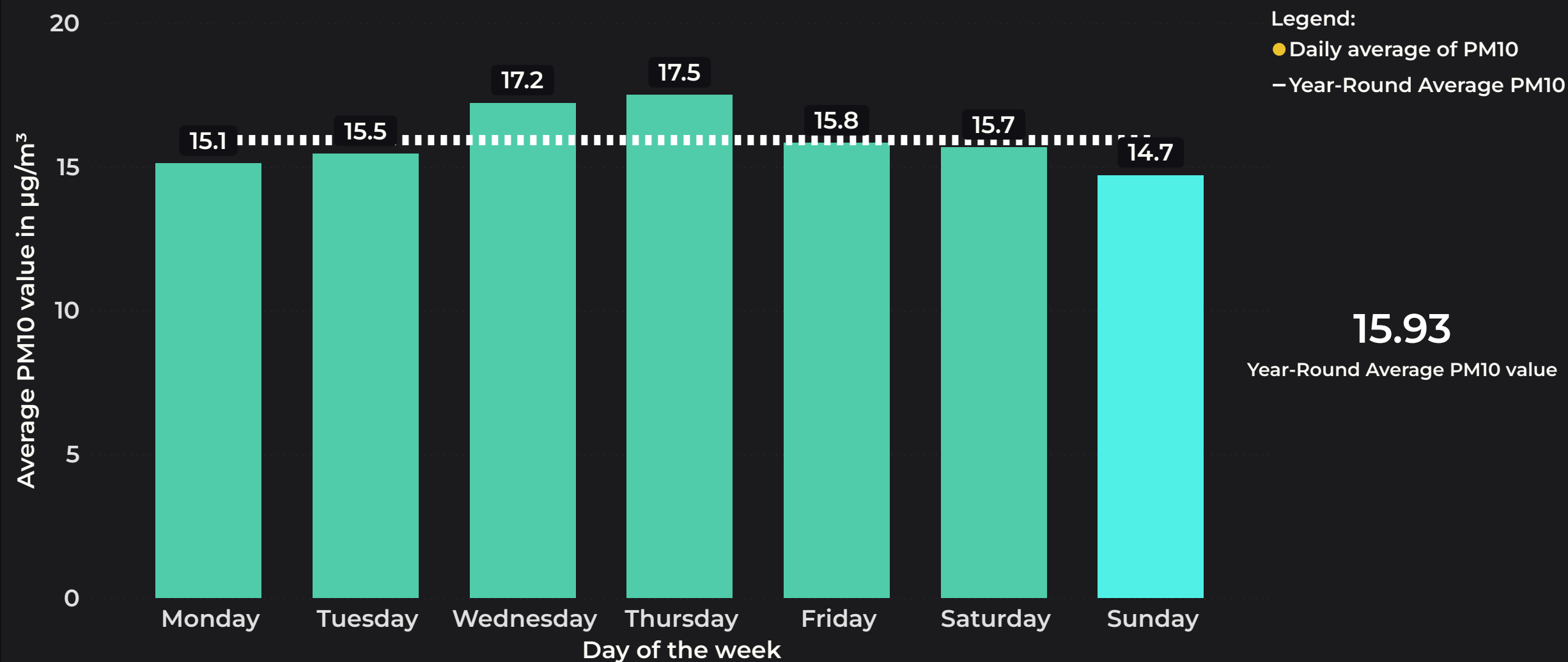
Answer:

The pattern shown is likely caused by the daily balance between **emissions** and **atmospheric dispersion**. During the afternoon, solar heating increases vertical air mixing, allowing pollutants to disperse through a deeper layer of air. In the evening, after sunset, the atmosphere becomes more stable and vertical mixing weakens. As a result, pollutants remain **closer to ground level** and **can accumulate more easily**.

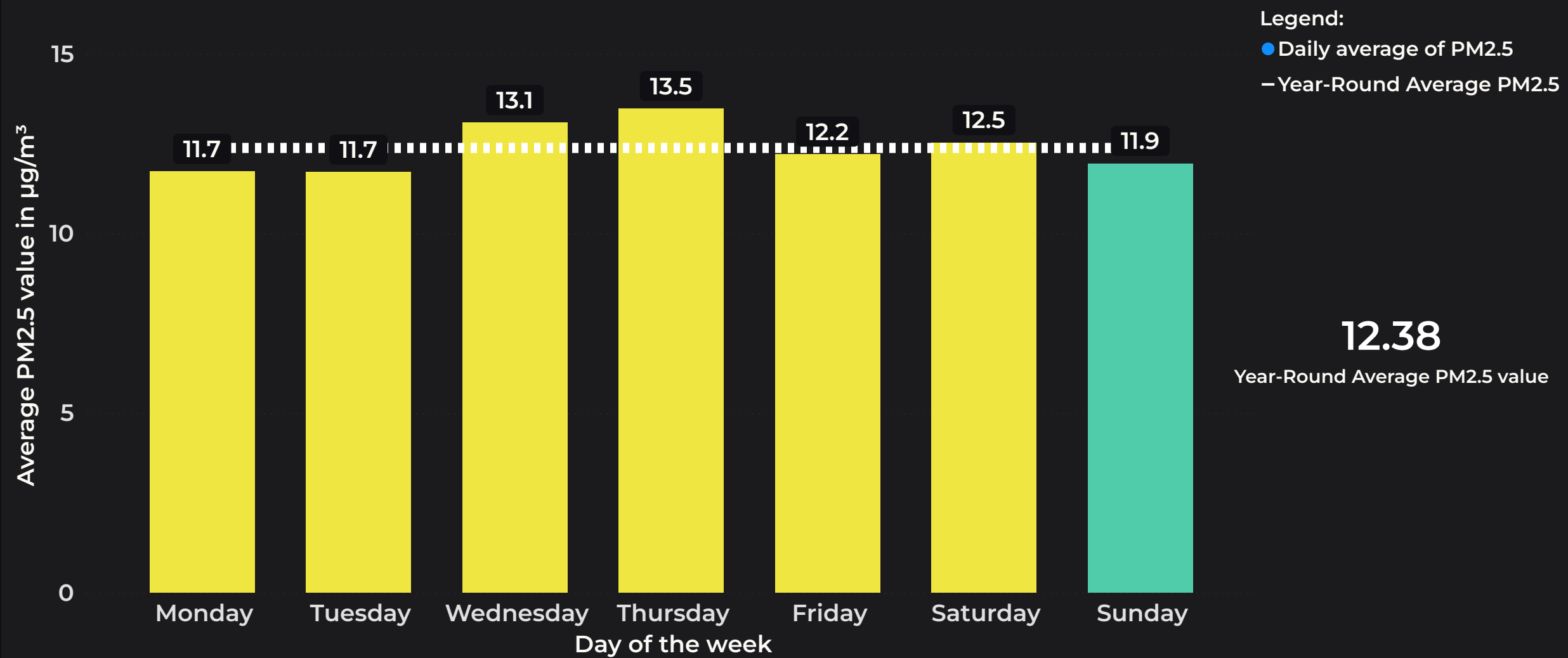
Evening traffic, residential heating, cooking and **other household activity** may continue to add emissions during this period. **Temperature inversion** can strengthen this effect, especially during colder months, by trapping cooler polluted air near the surface under a warmer layer of air.

Since the visual aggregates data across a full year, the result should be interpreted as a general daily pattern rather than proof of a single cause. **The 21:00–22:00 peak** is consistent with **weaker evening atmospheric dispersion, possible temperature inversions** and evening emissions. **The 15:00–16:00 minimum** is consistent with **stronger afternoon mixing**, when pollutants are more easily diluted. Confirming the exact causes would require comparing the hourly pollution values with meteorological data, season and node locations.

During which day of the week is air pollution highest throughout the city?



During which day of the week is air pollution highest throughout the city?



Answer:

The pattern shown is likely connected to regular weekday activity across the city. Mid-week days usually represent a stable working pattern, with **commuting, school traffic, commercial activity, deliveries, shops, markets** and **services** operating at normal intensity. These activities increase road traffic and congestion, which can contribute to both **exhaust emissions** and **non-exhaust particulate matter** such as **brake wear, tire wear, and road dust resuspension**.

The lower values on **Sunday**, and partly on **Monday**, also support this interpretation. The weekend usually has **reduced commuting, fewer deliveries, less construction activity** and **lower overall urban movement**. Monday may still be affected by the lower activity of the previous day or by a slower start to the working week.

I've considered **factory shifts** as a probable cause, but they are less likely to fully explain the weekly pattern, because many industrial operations **run continuously or on rotating shifts**.

Overall, the **Wednesday** and **Thursday peaks** should be interpreted as a **probable weekday activity effect** rather than the result of one single source. Confirming the exact causes would require comparing the weekday pollution values with traffic counts, meteorological data, holidays, school schedules, weekday/weekend patterns and node-level location patterns.

When and where were the absolute highest pollution readings recorded?

Year	Month	Day	Hour	Node location	PM10 value in $\mu\text{g}/\text{m}^3$
2025	March	15	16:00	Bulevardul Liviu Rebreanu	474.40
2025	March	15	22:00	Bulevardul Liviu Rebreanu	477.10
2025	March	15	17:00	Bulevardul Liviu Rebreanu	480.60
2025	March	15	23:00	Bulevardul Liviu Rebreanu	499.60
2025	March	13	21:00	Bulevardul Liviu Rebreanu	504.50
2025	March	15	12:00	Bulevardul Liviu Rebreanu	505.10
2025	March	15	09:00	Bulevardul Liviu Rebreanu	506.00
2025	March	13	23:00	Bulevardul Liviu Rebreanu	509.20
2025	March	16	00:00	Bulevardul Liviu Rebreanu	519.00
2025	March	13	18:00	Bulevardul Liviu Rebreanu	521.70
2025	March	15	10:00	Bulevardul Liviu Rebreanu	559.40
2025	March	13	19:00	Bulevardul Liviu Rebreanu	593.10
2025	March	13	16:00	Bulevardul Liviu Rebreanu	601.50
2025	March	13	20:00	Bulevardul Liviu Rebreanu	604.40
2025	March	13	15:00	Bulevardul Liviu Rebreanu	806.20

When and where were the absolute highest pollution readings recorded?

Year	Month	Day	Hour	Node location	PM2.5 value in $\mu\text{g}/\text{m}^3$
2025	March	13	22:00	Bulevardul Liviu Rebreanu	456.30
2025	March	15	16:00	Bulevardul Liviu Rebreanu	465.40
2025	March	15	17:00	Bulevardul Liviu Rebreanu	469.80
2025	March	15	22:00	Bulevardul Liviu Rebreanu	473.40
2025	March	15	12:00	Bulevardul Liviu Rebreanu	474.60
2025	March	15	09:00	Bulevardul Liviu Rebreanu	475.80
2025	March	13	18:00	Bulevardul Liviu Rebreanu	494.00
2025	March	15	23:00	Bulevardul Liviu Rebreanu	496.50
2025	March	13	21:00	Bulevardul Liviu Rebreanu	501.10
2025	March	13	23:00	Bulevardul Liviu Rebreanu	504.40
2025	March	16	00:00	Bulevardul Liviu Rebreanu	514.80
2025	March	13	16:00	Bulevardul Liviu Rebreanu	527.40
2025	March	15	10:00	Bulevardul Liviu Rebreanu	542.20
2025	March	13	19:00	Bulevardul Liviu Rebreanu	582.40
2025	March	13	20:00	Bulevardul Liviu Rebreanu	582.70

Answer:

The **absolute highest** pollution readings were recorded at the **same node** during a condensed **one-week period in mid-March**. These readings were **hourly spikes** rather than **sustained 24-hour averages**.

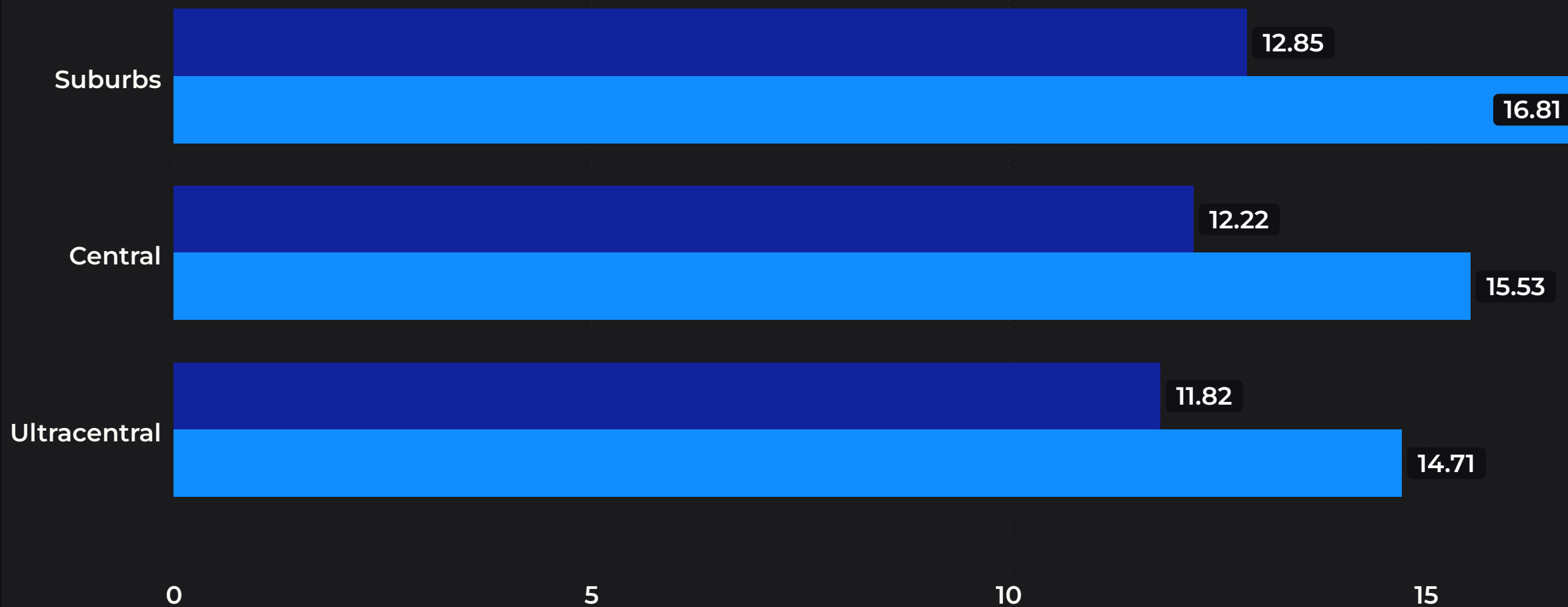
I've considered a regional explanation, such as **Saharan dust**, because it matches the time period when local authorities warned of a Saharan dust transport episode was reported over Romania in **mid-March 2025**.

Ultimately, I think it is unlikely to be the main cause. If Saharan dust were responsible, elevated PM10 values would normally be expected **across multiple nodes**, not only at **one sensor**. Instead, the pattern suggests either a very local pollution source near that node or a sensor-related issue. Possible local causes could include **nearby construction, road dust, localized burning, a chimney or exhaust source, heavy vehicle activity, or industrial-yard activity**. Alternatively, the sensor may have been affected by **calibration drift, contamination, humidity sensitivity, placement issues, or temporary malfunction**.

Therefore, these maximum readings are best treated as extreme node-specific values. They are useful for identifying an **anomaly** or **localized hotspot**, but they should be analyzed separately from the broader **citywide pollution pattern**. Confirming their cause would require checking whether nearby nodes recorded similar spikes, comparing the values with official monitoring stations, reviewing meteorological conditions, and looking for local events such as fires, roadworks, or dust-generating activity near the sensor.

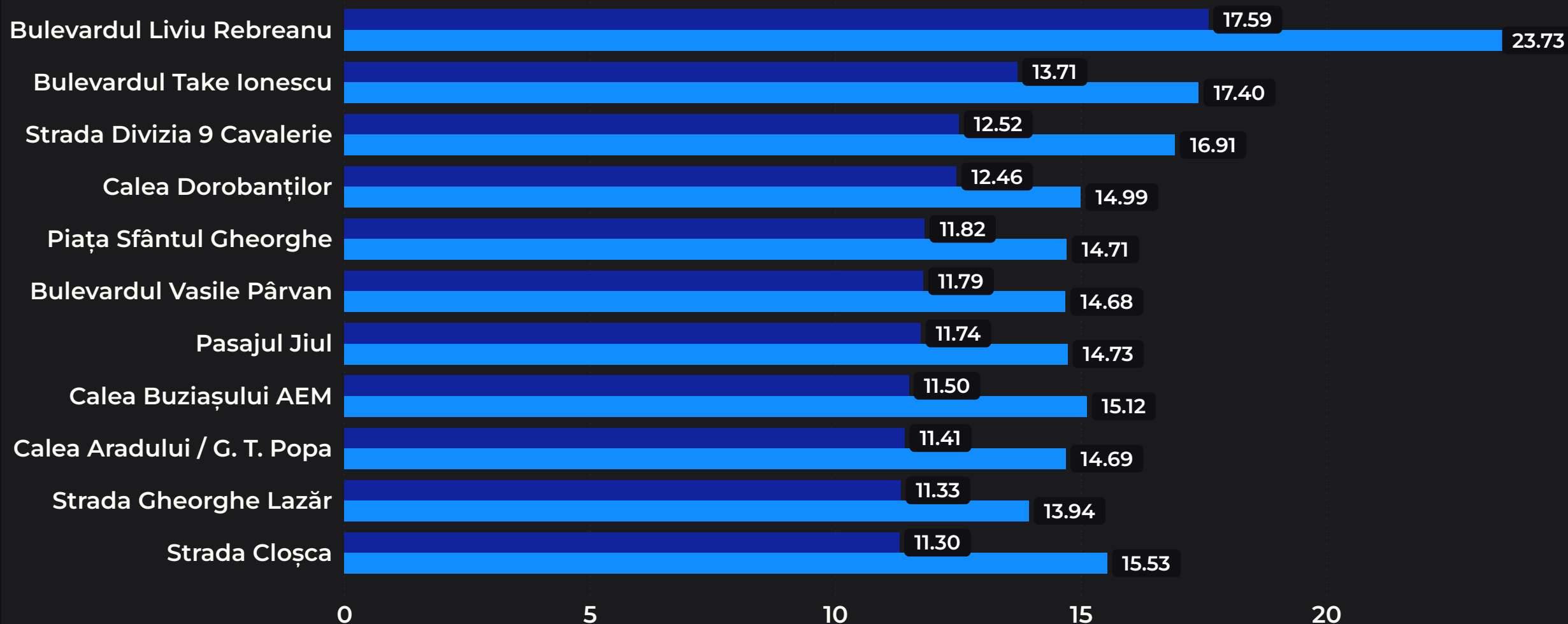
Which zone registers the highest year-round pollution?

● Average PM2.5 value per zone in $\mu\text{g}/\text{m}^3$ ● Average PM10 value per zone in $\mu\text{g}/\text{m}^3$



Which node registers the highest year-round pollution?

● Average PM2.5 value per node in μm ● Average PM10 value per node in μm



Answer:

The year-round pollution differences between the main zones are **relatively small**, suggesting that pollution is **not strongly concentrated** in **only one broad area** of the city. For **PM2.5**, comparing the lowest zone average of to the highest zone average shows that the gap between zones is limited.

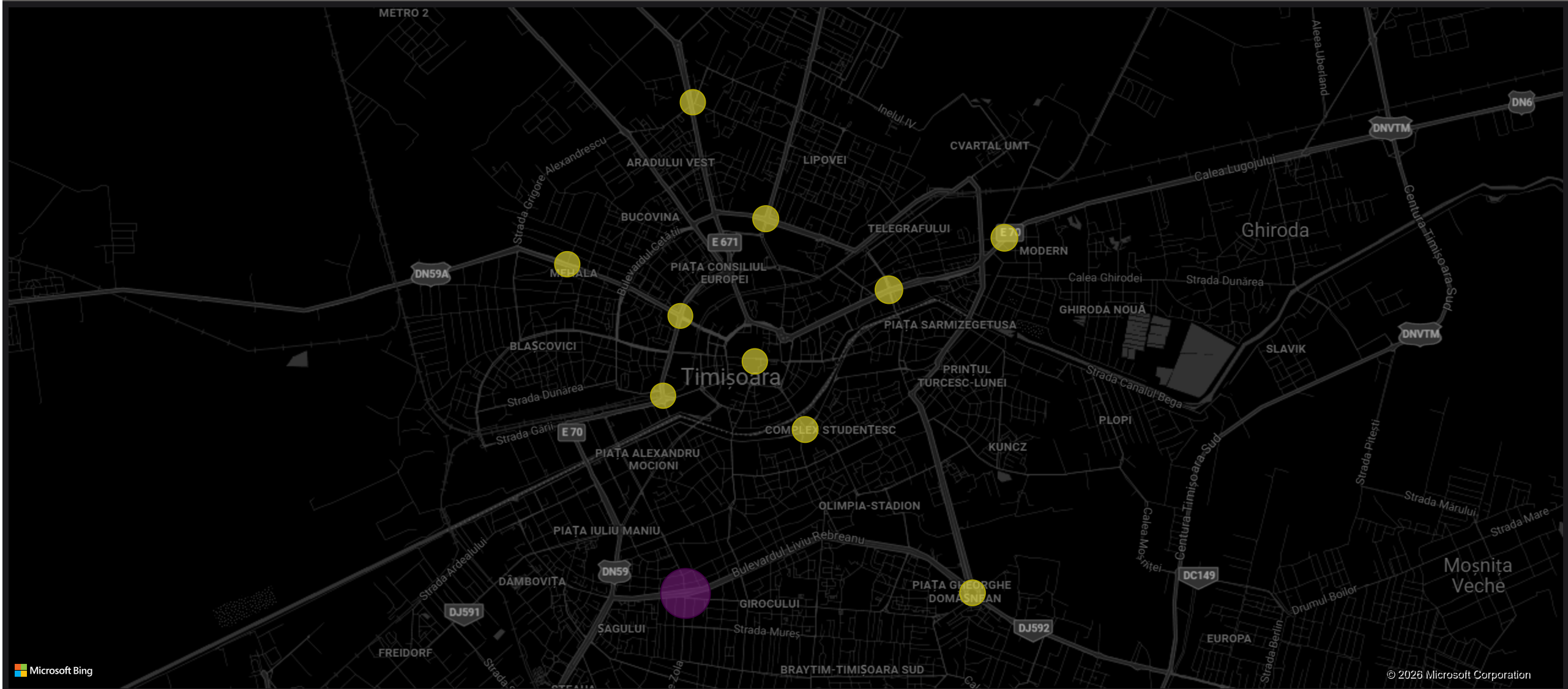
This result is not unexpected, since the ultracentral area contains **more pedestrian spaces**, while the suburban zone includes more **roads, factories, heating plants, electric plants**, and other **pollution-related infrastructure**.

However, the **node-level results** show a clearer difference. The same node that recorded the **extreme pollution spikes** also has the **highest year-round average** values, This suggests that the node may be located near a **persistent local pollution source**. At the same time, because this is also the node responsible for the **extreme spike readings**, the possibility of a sensor anomaly, calibration issue, or unrepresentative placement should also be considered.

Therefore, the zone-level result points to a **relatively balanced** citywide distribution, while the node-level result highlights a specific location that behaves differently from the rest of the network.

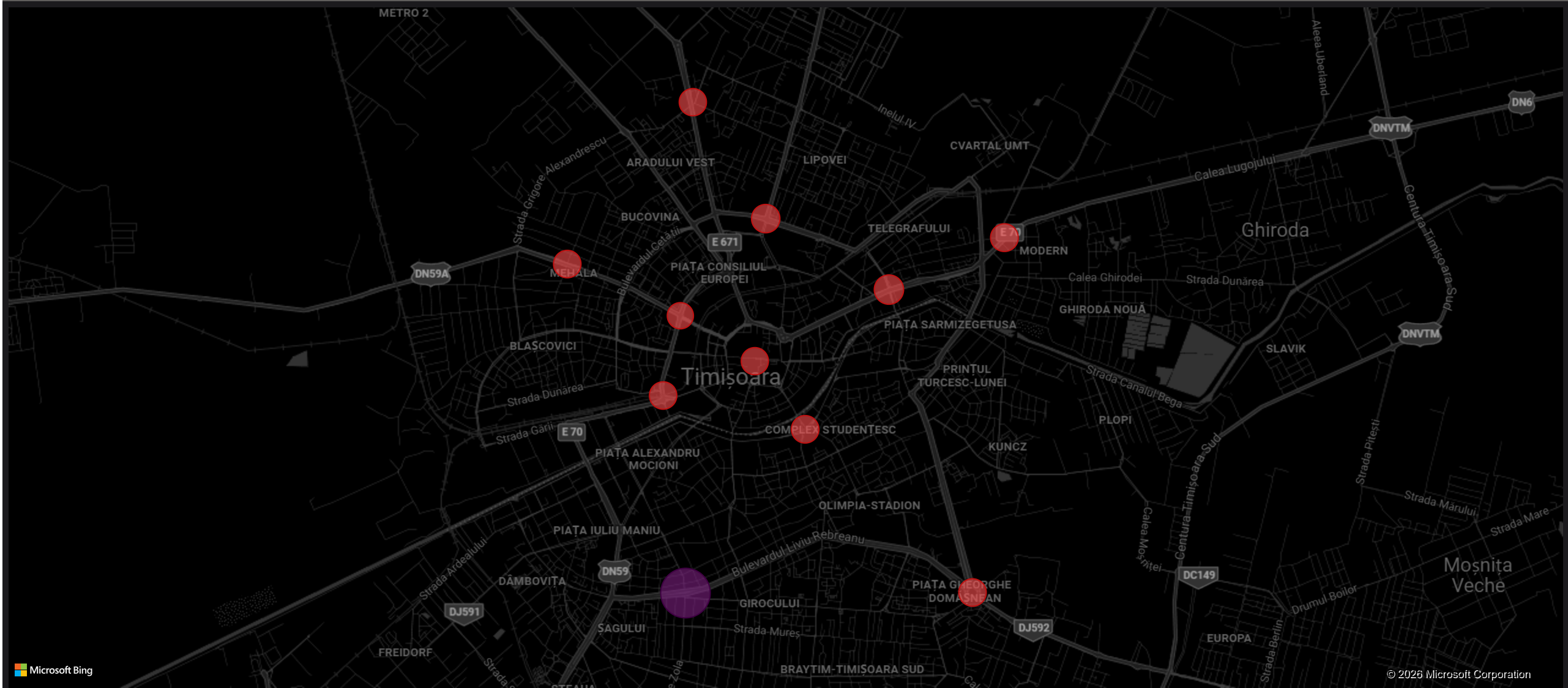
Which node shows the highest largest pollution variation?

P99 – P1 PM10 variation between nodes (Larger bubbles indicate nodes where pollution levels fluctuate more strongly across the year.)



Which node shows the highest largest pollution variation?

P99 – P1 PM2.5 variation between nodes (Larger bubbles indicate nodes where pollution levels fluctuate more strongly across the year.)



Answer:

The map visual uses the **P1–P99 range** to compare pollution variation between sensor nodes. This metric shows the spread between very low and very high hourly values while excluding the most extreme **1%** of readings at both ends. It was chosen because the **P5–P95** range made the bubble-size differences **difficult to notice**, while using **raw maximum** values caused the **March peak node (Bulevardul Liviu Rebreanu)** to dominate the map too strongly.

This visual should be interpreted as a **measure of pollution fluctuation**, not average pollution. A **larger bubble** indicates a node where **pollution levels vary more strongly** throughout the year, either because of intermittent local sources, short-term pollution episodes, changing weather conditions, poor dispersion, or possible sensor instability.

For **PM10**, most nodes have a similar **P1–P99** range, roughly from **1 to 60 $\mu\text{g}/\text{m}^3$** . For **PM2.5**, most nodes have upper **P99** values around **70–80 $\mu\text{g}/\text{m}^3$** , suggesting that fine-particle pollution episodes are more **broadly distributed** across the city. In both cases, the aforementioned node stands out as an outlier.

Sources:

A1:

<https://www.airdata.ro/map>

A2:

<https://aaqr.org/articles/aaqr-21-10-0a-0294>

<https://pubs.acs.org/doi/10.1021/acs.estlett.8b00573>

<https://www.eea.europa.eu/en/analysis/maps-and-charts/temperature-inversion-traps-pollution-at>

A3:

<https://www.sciencedirect.com/science/article/abs/pii/S1352231011005851>

A4:

[https://www.meteoromania.ro/wp-](https://www.meteoromania.ro/wp-content/uploads/comunicate/Comunicat_praf_saharian_09_martie_2025.pdf)

[content/uploads/comunicate/Comunicat_praf_saharian_09_martie_2025.pdf](https://www.meteoromania.ro/wp-content/uploads/comunicate/Comunicat_praf_saharian_09_martie_2025.pdf)

Q5, Q6 & Q7:

Data from sources and visuals